

**Course: Software Engineering Lab (UE23CS352A)**

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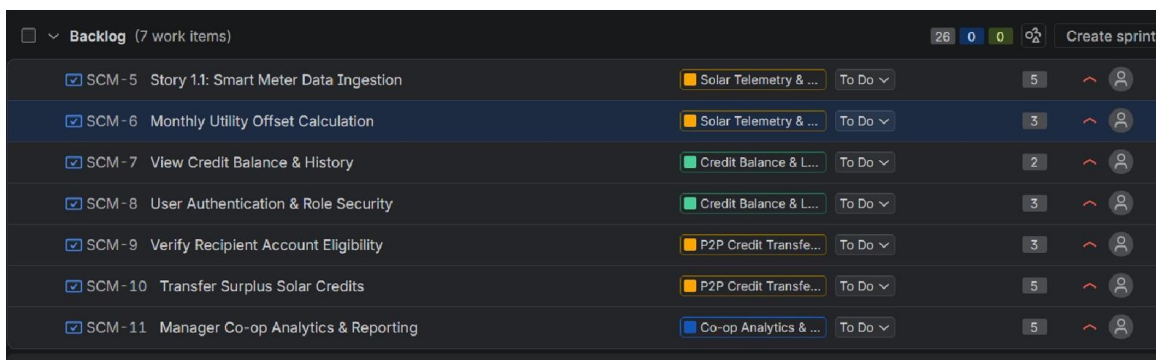
**SRN: PES1UG24CS117**

**Section: 5B**

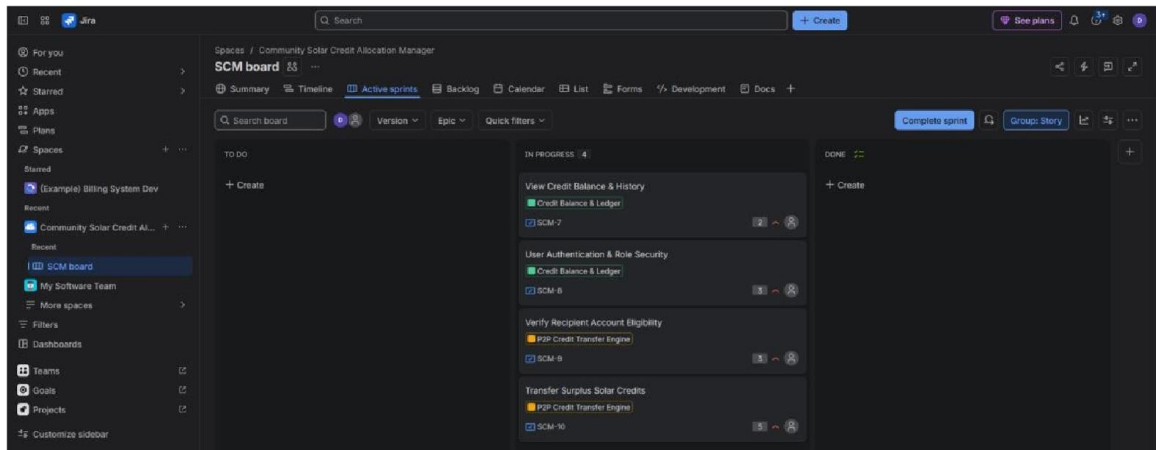
## Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Project: Community Solar Credit Allocation Manager

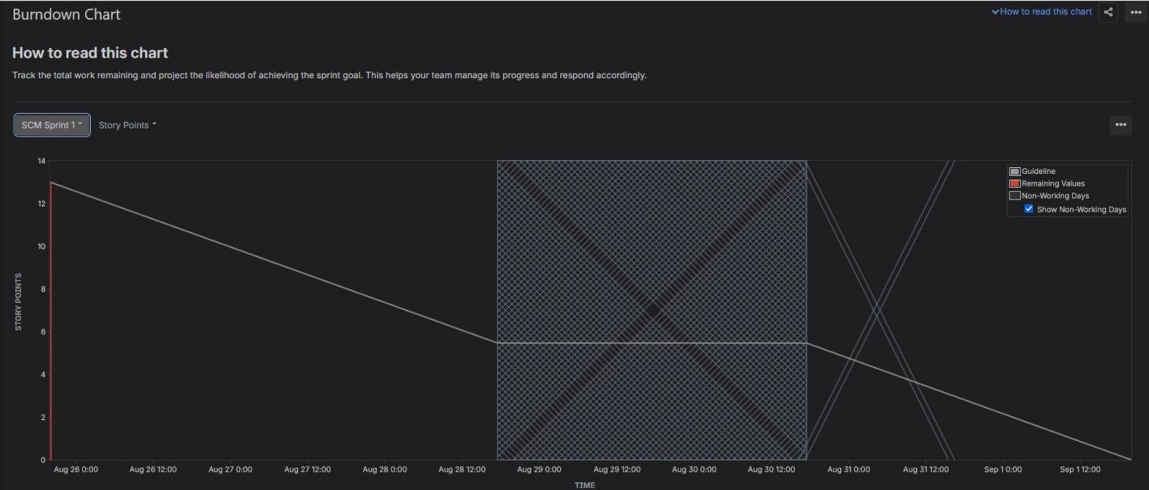
**Jira Project/Workspace Link:** <https://dineshkumarc2006-1787648279359.atlassian.net/jira/software/c/projects/SCM/boards/3>



|                                     |                                             |                       |         |   |    |               |
|-------------------------------------|---------------------------------------------|-----------------------|---------|---|----|---------------|
| <input type="checkbox"/>            | ▼ Backlog (7 work items)                    | 26                    | 0       | 0 | 🔍  | Create sprint |
| <input checked="" type="checkbox"/> | SCM-5 Story 1.1: Smart Meter Data Ingestion | Solar Telemetry & ... | To Do ▼ | 5 | ⬇️ | 👤             |
| <input checked="" type="checkbox"/> | SCM-6 Monthly Utility Offset Calculation    | Solar Telemetry & ... | To Do ▼ | 3 | ⬇️ | 👤             |
| <input checked="" type="checkbox"/> | SCM-7 View Credit Balance & History         | Credit Balance & L... | To Do ▼ | 2 | ⬇️ | 👤             |
| <input checked="" type="checkbox"/> | SCM-8 User Authentication & Role Security   | Credit Balance & L... | To Do ▼ | 3 | ⬇️ | 👤             |
| <input checked="" type="checkbox"/> | SCM-9 Verify Recipient Account Eligibility  | P2P Credit Transfe... | To Do ▼ | 3 | ⬇️ | 👤             |
| <input checked="" type="checkbox"/> | SCM-10 Transfer Surplus Solar Credits       | P2P Credit Transfe... | To Do ▼ | 5 | ⬇️ | 👤             |
| <input checked="" type="checkbox"/> | SCM-11 Manager Co-op Analytics & Reporting  | Co-op Analytics & ... | To Do ▼ | 5 | ⬇️ | 👤             |



|                                                                              |  |                                      |          |           |   |    |   |
|------------------------------------------------------------------------------|--|--------------------------------------|----------|-----------|---|----|---|
| Jira                                                                         |  | Search                               | + Create | See plans | 🔔 | ⚙️ | 👤 |
| Spaces / Community Solar Credit Allocation Manager                           |  |                                      |          |           |   |    |   |
| SCM board                                                                    |  |                                      |          |           |   |    |   |
| Summary Timeline Active sprints Backlog Calendar List Forms Development Docs |  |                                      |          |           |   |    |   |
| Search board Version Epic Quick filters                                      |  |                                      |          |           |   |    |   |
| TO DO                                                                        |  | IN PROGRESS (4)                      |          | DONE      |   |    |   |
| + Create                                                                     |  | View Credit Balance & History        |          | + Create  |   |    |   |
|                                                                              |  | Credit Balance & Ledger              |          |           |   |    |   |
|                                                                              |  | SCM-7                                |          |           |   |    |   |
|                                                                              |  | User Authentication & Role Security  |          |           |   |    |   |
|                                                                              |  | Credit Balance & Ledger              |          |           |   |    |   |
|                                                                              |  | SCM-8                                |          |           |   |    |   |
|                                                                              |  | Verify Recipient Account Eligibility |          |           |   |    |   |
|                                                                              |  | P2P Credit Transfer Engine           |          |           |   |    |   |
|                                                                              |  | SCM-9                                |          |           |   |    |   |
|                                                                              |  | Transfer Surplus Solar Credits       |          |           |   |    |   |
|                                                                              |  | P2P Credit Transfer Engine           |          |           |   |    |   |
|                                                                              |  | SCM-10                               |          |           |   |    |   |



| TIME              |        |                              |                 |              |      |           |
|-------------------|--------|------------------------------|-----------------|--------------|------|-----------|
| Date              | Issue  | Event Type                   | Event Detail    | Story Points |      |           |
|                   |        |                              |                 | Inc.         | Dec. | Remaining |
| 25/Aug/26 8:07 PM | SCM-11 | Sprint start                 |                 | 5            |      |           |
|                   | SCM-5  |                              |                 | 5            |      |           |
|                   | SCM-6  |                              |                 | 3            |      |           |
|                   | Total  |                              |                 |              |      | 13        |
| 25/Aug/26 8:09 PM | SCM-5  | Burndown                     | Issue completed |              | 5    | 8         |
|                   | SCM-6  | Burndown                     | Issue completed |              | 3    | 5         |
|                   | SCM-11 | Burndown                     | Issue completed |              | 5    | 0         |
|                   | SCM-11 | Sprint ended by dineshkumarc |                 |              |      | 0         |
|                   | SCM-5  |                              |                 |              |      | 0         |
|                   | SCM-6  |                              |                 |              |      | 0         |
|                   | Total  |                              |                 |              |      | 0         |

## Reflection Questions & Answers

### Q1: Did your estimations reflect the actual effort?

Answer: Yes, the estimations successfully reflected the complexity and risk associated with each task. Integration-heavy features such as Smart Meter Telemetry Ingestion (SCM-5) and P2P Credit Transfers (SCM-10) were assigned 5 story points due to handling external APIs, ledger locking, and transaction consistency. Simpler CRUD or read-only tasks like viewing credit balances (SCM-7) were estimated at 2 story points, aligning the non-linear Fibonacci progression with actual effort differences.

### Q2: Was your backlog well-prioritized?

Answer: Yes, the backlog was prioritized using a value-first approach. Core user security, data validation, and the transaction engine (Sprint 1) were completed first, as they represent the Minimum Viable Product (MVP) requirements. Data ingestion pipelines and managerial reports (Sprint 2) were scheduled next, ensuring dependencies were resolved sequentially.

### Q3: How did your simulated sprint align with your plan?

Answer: The simulated sprint aligned precisely with our target velocity of 13 story points per sprint. There was no scope creep during active sprints, and all committed items were successfully delivered within the timebox.

### Q4: What insights did the burndown chart give about your team's capacity?

Answer: The burndown chart illustrated a steep drop-off of story points as work items were finalized. In real-world team sprints, a flatter step-down indicates continuous, incremental completion of tasks, while late burndowns suggest that work was either underestimated, blocked, or not adequately broken down into smaller sub-tasks.